Statistical Sample Selection and Multivariate Knowledge Mining for Lightweight Detectors in Remote Sensing Imagery

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***Abstract*— In recent years, more concerns are shed on the lightweight detection model in remote sensing (RS), but it is difficult to reach a competitive performance relative to the deep model. Knowledge distillation has been verified as a promising method, which can promote the performance of the lightweight model without extra parameters. While there are two key issues of detection distillation, one is the sample selection and the other is the knowledge selection. Since the varying object size and complex features in RS, the existing methods based on the fixed threshold are incapable of selecting the optimal distillation samples and they also ignore the potential multivariate knowledge among RS samples simultaneously. In this article, we propose a statistical sample selection and multivariate knowledge mining framework. The statistical sample selection module formulates the task as the modeling and splitting of the probability distribution of sample selection cost, which is more suitable for dynamically choosing multiscale samples in RS and eliminates the distortion of previous static distillation selection. Furthermore, to mine the complex feature knowledge of samples in RS, we design a multivariate knowledge mining module, in which knowledge includes explicit and implicit knowledge. The proposed module validly delivers the core knowledge from the teacher model to the lightweight model. Massive experiments on three challenging RS datasets [a large-scale Dataset for Object deTection in Aerial images (DOTA), Northwestern Polytechnical University very-high-resolution 10-class (NWPU VHR-10), and object DetectIon in Optical Remote sensing images (DIOR)] prove that our method achieves state-of-the-art performance.**

***Index Terms*— Knowledge distillation (KD), lightweight model, object detection, remote sensing (RS) imagery.**

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I. Introduction

THE rapid development of remote sensing (RS) technology brings many optical aerial images, while detectors [1]–[4] based on the Deep Convolutional Neural Network (DCNN) can validly detect objects in RS images. Although these detectors have high accuracy, they usually consume much storage, have a complicated structure, and need more calculation resources, which result in the difficulty of intelligent terminal deployment (drones and high-altitude satellites). Therefore, the lightweight model has become key research in the RS realm recently.

Some researchers have proposed many lightweight models [5]–[8] for the object detection task. However, since the small capacity of the backbone, the lightweight model usually cannot extract the feature of objects effectively, which makes the subsequent detection imprecise, including false alarm, misclassification, and positional deviation.

Inspired by the knowledge distillation (KD) proposed by Hinton *et al.* [9], a cumbersome teacher model can teach a lightweight student model dark knowledge besides the ground truth, which facilitates the student model to mining more potential information from the dataset. Hence, KD is a promising solution for the optimization of the lightweight model. The existing KD [10]–[13] methods are usually suitable for the classification task, but the imbalance between foreground and background in the detection task results in limited promotion for performance when directly migrating these methods from classification to detection task. Therefore, some researchers [14]–[16] propose to only choose the core part of the input images for distillation to avoid the noise of complex background.

However, the above-mentioned approaches still have two issues in RS.

A. Static Sample Selection

For the KD of the detection task, there is a key issue, which sample is worth to be selected. The sample represents a potential object in the input image. Existing works usually adopt the following approaches to decide on samples. Mimick [14] proposes that the RoI regions contain key information. Distilling Object Detectors with Fine-grained Feature Imitation (FGFI) [15] considers the regions, which are close to the

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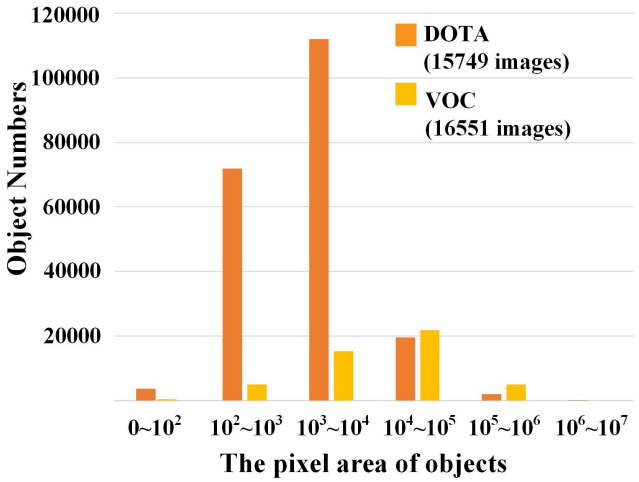

Fig. 1. Object pixel area statistics. We count the pixel area of each object in the training dataset. Compared with the natural scene dataset [Visual Object Class (VOC)], the sizes of objects in the RS dataset (DOTA) are more discrete and have more objects in fewer number training images.

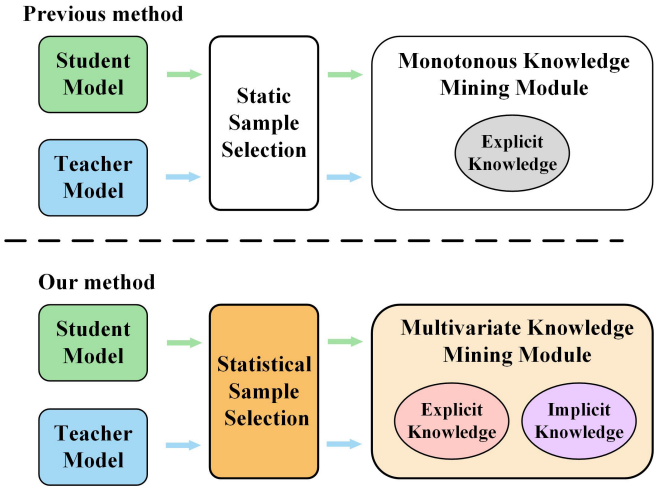

Fig. 2. Illustration of the task framework. The previous method statically selects samples and performs monotonous knowledge mining. Our method dynamically selects statistical samples and conducts multivariate knowledge mining.

object in the feature map of the teacher model, are the core position. Furthermore, Distilling Object Detectors with Task Adaptive Regularization (TAR) [16] puts forward the ground truth region as the crux. In summary, their sample selection methods usually need well-designed hyperparameters, such as IoU threshold and ground truth region. However, RS images usually contain multiscale objects (see Fig. 1) and the fixed IoU threshold is not suitable for multiscale objects simultaneously. Besides, small objects take a small proportion in feature maps and the mapping of the ground truth region will result in regional misalignment. Moreover, whether those methods use IoU threshold or ground truth, they just consider the location information as the criterion, while the category information is ignored. The optimal distillation sample will benefit from applying the rich category information of RS objects.

B. Monotonous Knowledge Mining

The second problem of the KD in the detection task is which knowledge should be delivered. FGFI [15] considers the feature of objects as knowledge. Mimick [14] and TAR [16] add the classification probability and regression location as extra knowledge. In general, they focus on monotonous knowledge mining, which is defined as transferring the outputs of the corresponding sample between two models. While the potential relation information of sample groups is neglected, more samples are provided by massive objects in RS images (See Fig. 1), so more relations including category and geometry between samples can be mined, which is valuable for lightweight model optimization.

Considering the above-mentioned issues of existing KD methods of the detection task in the RS realm, we propose a statistical sample selection and multivariate knowledge mining framework, which is presented in Fig. 2. We want to elegantly select the optimal distillation samples for multiscale RS objects without the fixed hand-craft hyperparameters and synchronize with the training phase of the student model. The statistical sample selection module proposed by us formulates the task as modeling the probability distribution of sample selection cost, including the location and category information. Then, a low-cost split scheme is adopted to choose the optimal distillation samples from the probability distribution. A secondary sample screening is performed to obtain the independent classification and regression distillation set by the cost comparison between the teacher and student models. Furthermore, to eliminate monotonous knowledge mining brought the issue of underutilized information among massive samples in RS images, we put forward the multivariate knowledge mining method, which contains implicit and explicit knowledge. The implicit knowledge delivers the sample pairs' potential category relation and the group discreteness of geometry. And the explicit knowledge transfers the feature, classification, and regression multi-attribute information of the object. Hence, the lightweight student model can benefit from the multivariate information provided by the teacher model, thereby optimizing the model better.

The main innovations and contributions of this article are summarized as follows.

1) We design a general distillation framework (S^3MKM) for lightweight detector optimization in the RS realm. Massive experiments on a large-scale Dataset for Object deTection in Aerial images (DOTA), Northwestern Polytechnical University very-high-resolution 10-class (NWPU VHR-10), and object DetectIon in Optical Remote sensing images (DIOR) datasets prove our effectiveness, which exceeds other SOTA distillation approaches. Besides, the performance of the lightweight model optimized by our S^3MKM transcends classic detectors.
2) The statistical sample selection module proposed by us can dynamically determine the optimal samples for distillation, which models the sample selection cost as a probability distribution and maximizes its likelihood by the expectation–maximization (EM) algorithm. The optimal distillation samples are obtained from the probability distribution by a low-cost split scheme.

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3) We propose a multivariate knowledge mining method, which contains implicit knowledge mining (IKM) and explicit knowledge mining (EKM). IKM focuses more on the potential relation between samples. While EKM pays attention to corresponding sample direct matches.

We arrange the residual article as below. The related works are presented in Section II. Our method is particularly elaborated in Section III. We present and analyze the results of experiments in Section IV. We sum up our work in Section V finally.

II. Related Works

A. Lightweight Object Detection

The existing object detectors [1]–[4], [17]–[20] require massive computation resources and storage, which impede them to be deployed on intelligent terminals with limited computing power. In general, they usually adopt a heavy backbone, including VGG16 [21] and ResNet [22], which makes the detectors cumbersome. Hence, more light is shed on the lightweight backbone design. The depthwise separable convolutions proposed in the MobileNet [5] make the deep neural networks lightweight. Many lightweight detectors, like MobileNet-SSD [5] and MobileNetV2-SSDLite [6], are designed when MobileNet is combined with one-stage detectors.

The previous detectors usually migrate the backbone designed for the image classification task [21], [22] to the detection task. However, the different characteristics between classification and detection tasks result in the lightweight backbone is not fit to directly migrate. Therefore, the Tiny-DSOD [23] and the Pelee [7] propose to design a unique lightweight backbone for the detection task. Besides, ThunderNet [8] designs a compression backbone, Region Proposal Network (RPN), and detection head to reduce the computation. It achieves real-time detection on ARM platforms finally.

Different from the natural images, the RS images usually have multiscale objects and a complex background, which makes the above-mentioned lightweight detectors difficult to achieve effective detection. Hence, some researchers propose many deep-learning-based detection approaches for RS images. Wang *et al.* [24] adopt the area-weighted loss and ASFP module to fuse multiscale features and focus more on small objects. Sun *et al.* [25] put forward the context-based deformable module (CBD) to enhance the detection results of rigid objects by combining Deformable Convolution Network (DCN) and attention. Wei *et al.* [26] design the detection task as the selection and clustering of intersection lines paired inside the aircraft. Besides, the new pentagonal mask proposed by them can provide a more precise location. Zhu *et al.* [27] propose an automatic organization mechanism to detect rotating objects. Hu *et al.* [28] propose the multi-scale feature fusion SNET (MSF-SNET) which is a lightweight detector for RS detection.

B. Knowledge Distillation

KD can be regarded as an information transfer process and the teacher model will provide potential knowledge to help the student mine the key information of data. Many researchers have proposed various kinds of approaches in recent years. Ways to realize the knowledge transfer can be mainly categorized into two main families.

Some researchers focus more on explicit KD. It usually transfers the knowledge of the same sample directly between the student and teacher models. Feature and classification knowledge are major mining information in the task of classification. Hinton *et al.* [9] first propose the KD for classification tasks. Usually, a teacher model is used to deliver dark information, which is the soft label of the output from the teacher model, to the student network. FitNet [10] introduces "hint learning" to distill the student model, where the hint is the information of the teacher's intermediate layers. Attention (AT) [11] proposes that student mimics the attention maps of the teacher. Although there is a lot of research on the application of KD to classification tasks, it is difficult to directly migrate to more complicated detection tasks. More works devote themself to the selection of object knowledge. And except for the classification and feature knowledge, regression knowledge is added for better optimization in the detection task. Learning Efficient Object Detection Models with Knowledge Distillation (LEOD) [29] first proposes to use KD in object detection and the student model imitates each knowledge of the teacher model. Mimick [14] proposes the student model to imitate the RoI region in the feature map of the teacher model, which contains negative and positive samples. FGFI [15] suggests that the student model only imitates the regions close to the object in the teacher model's feature map. TAR [16] proposes that the student model imitates the teacher model's feature map of the ground truth region, and only imitates the output result of the positive sample in the classification and regression heads. DeFeat [30] elucidates that features from different regions should be assigned different importance during distillation. Adaptive Reinforcement Supervision Distillation (ARSD) [31] proposes an multiscale core features imitation (MCFI) module for selecting the multiscale core features imitation and an strict supervision regression distillation (SSRD) module for choosing the optimal location distillation.

Others pay attention to implicit KD. It usually transfers the relational knowledge of sample pairs between the student and teacher models. Relational Knowledge Distillation (RKD) [12] proposes a relational KD method that transfers the distance and angle knowledge of sample pairs. Yim *et al.* [13] propose a flow of solution process (FSP), which calculates the Gram matrix between two layers. Then, the student model imitates the FSP matrix of the teacher model. Double Similarity Distillation (DSD) [32] proposes a pixelwise and categorywise similarity distillation to deliver the knowledge of residual attention maps and correlation matrix.

III. Methods

A. Overview

The KD can be regarded as an information migration from the teacher to the student model. Given a sample set E from the input image x and the E represents all the potential objects in the input image. The information transfer is defined as follows:

$$\mathcal{S}(\Psi(\Upsilon(E))|\theta_s, x) \Leftarrow \mathcal{T}(\Psi(\Upsilon(E))|\theta_t, x) \tag{1}$$

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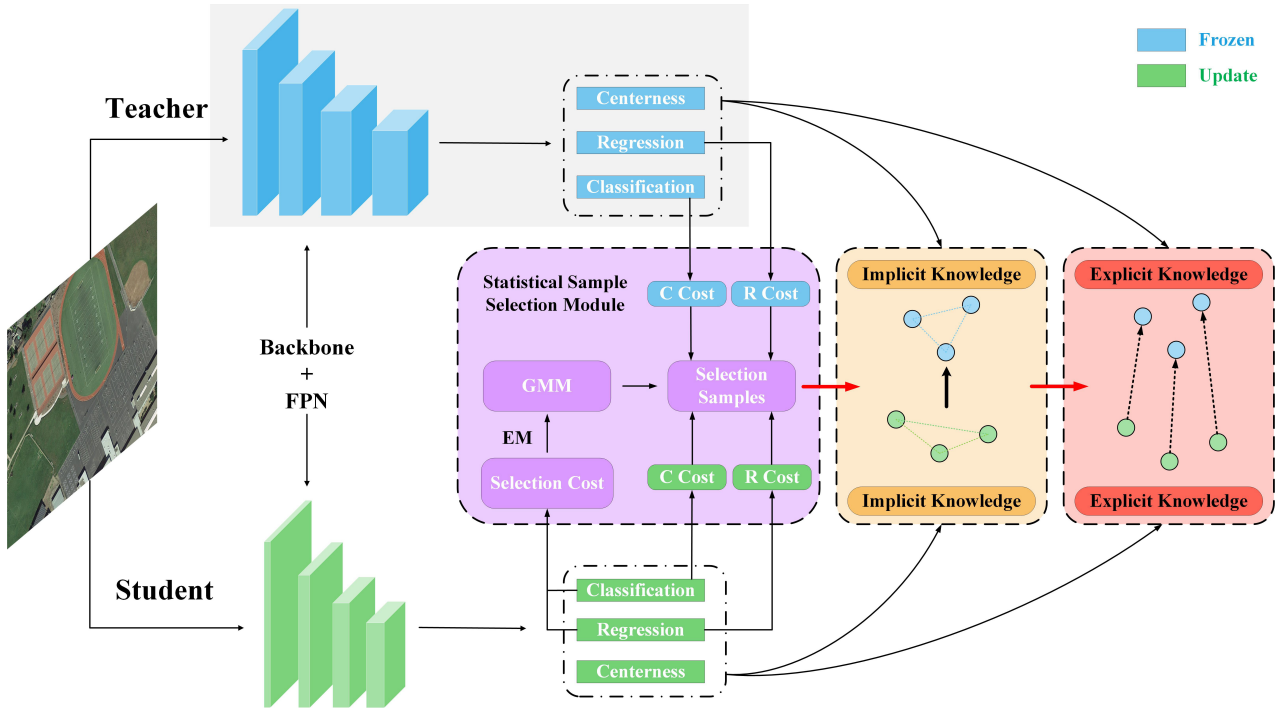

Fig. 3. Pipeline of the S^{3}MKM framework. The statistical sample selection module chooses the optimal samples for distillation. Samples are sent to the IKM module and EKM module. The IKM module pays attention to the relation between the sample pairs. The EKM module focuses on the imitation of corresponding sample knowledge between the student and teacher models. During the training phase, the parameters of the teacher model are frozen, while the student model is updated by the ground truth and the knowledge from the teacher model.

where θ_s and θ_t represent the parameters of the student model $\mathcal{S}$ and the teacher model $\mathcal{T}$, respectively. From (1), it can be observed that there are two key operations $\Upsilon(\bullet)$ and $\Psi(\bullet)$.

1) Sample Selection: The detection head usually generates massive samples but not all samples have the worth for distillation. The operation $\Upsilon(\bullet)$ means choosing the optimal sample of them for distillation.

2) Knowledge Mining: The other key operation is which knowledge of samples should be delivered. The operation $\Psi(\bullet)$ means mining the potential knowledge of samples.

Based on the above-mentioned theory, we propose a framework that tackles the issue of sample selection and knowledge mining for distillation in the RS realm. Fig. 3 provides a diagram of our framework. Both the teacher and student models adopt the classic single-stage detector, which includes the backbone network, feature pyramid structure, and detection head. Furthermore, the detection head contains three branches, such as classification, regression, and centerness. During the training phase, the teacher model is frozen but the student model is updated by potential knowledge of the teacher model and ground truth.

B. Statistical Sample Selection Module

Sample selection for distillation in detection is a core issue. Compared with the natural images, more multiscale objects are in RS images. And the size of objects varies extremely, which is not suitable for selecting distillation samples through the fixed threshold. Hence, we propose a statistical sample selection module for dynamically choosing that eliminates the selection distortion of previous methods.

Given an input image x, a sample e that belongs to it, the model f_θ that has parameters θ, and we can get the outputs of this sample as follows:

$$f_\theta(e, x) = \{c, r\} \tag{2}$$

where c and r are its predictions of classification and regression, respectively.

To evaluate the quality of the sample e, we adopt the selection cost as the index. The selection cost $\mathcal{C}(e)$ is calculated by classification and regression cost according to its corresponding ground truth gt as follows:

$$\mathcal{C}(e) = \mathcal{C}_{\text{cls}}(c, gt) + \omega \mathcal{C}_{\text{reg}}(r, gt) \tag{3}$$

where $\mathcal{C}_{\text{cls}}$ denotes the focal loss and $\mathcal{C}_{\text{reg}}$ is the GIoU loss. ω is a balancing coefficient to control the relation between two costs.

After getting the selection cost of all samples, we want to acquire the distillation samples for each ground truth from positive samples. Driven by the above-mentioned thought, we will model selection costs as probability distributions and separate them into two groups. One group is used for distillation and the other is useless. We adopt the Gaussian mixture model (GMM) to model the sample selection cost distribution by two modalities

$$P(e|gt, \theta, x) = \eta_1 \mathcal{D}_1(e; \nu_1) + \eta_2 \mathcal{D}_2(e; \nu_2) \tag{4}$$

where $\mathcal{D}_1$ and $\mathcal{D}_2$ represent two distributions of the selected samples and useless samples. η_1, η_2, ν_1, and ν_2 denote the weight and initialization parameters of two distributions,

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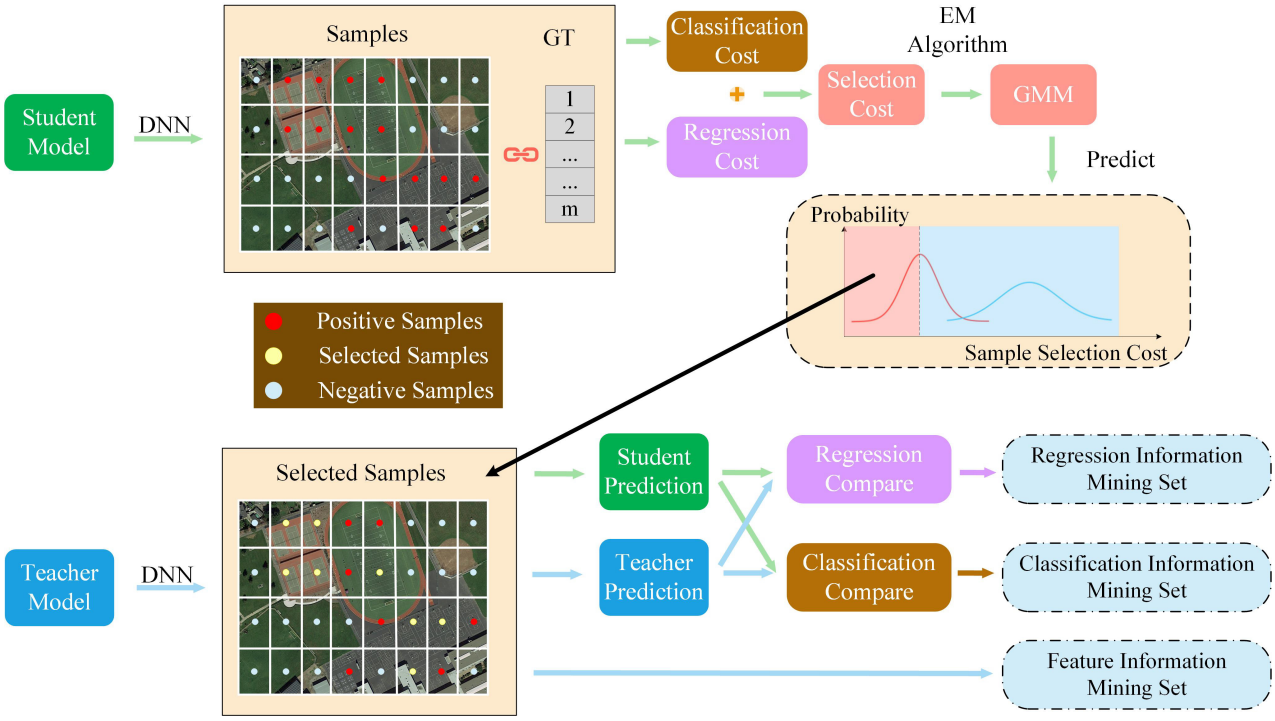

Fig. 4. Architecture of the statistical sample selection module. We can get the positive samples (red dots) for parameters updating from the student model. We calculate their selection cost according to the prediction and ground truth, then adopt the GMM to model the sample selection cost distribution by two modalities. The EM algorithm is used to optimize the likelihood of this GMM. The statistical samples (yellow dots) are selected by the low-cost distribution. Finally, we get three information mining sets according to the selection cost of the student and teacher.

respectively. Given a set of sample costs belonging to ground truth, we adopt the EM algorithm to optimize the likelihood of the GMM. Since we get the estimating GMM, we can obtain the probability of each sample whether it is selected or not. Then, a scheme is used to split samples into two groups by their probability values. We choose the low-cost group in the probability distribution as distillation samples (see Fig. 4). Loop over the above operation for each ground truth, we will get all the distillation samples. For the teacher model, we also get the same samples according to distillation sample indexes from the student model.

The set of selected samples $\mathcal{E}$ is obtained by the GMM and split scheme. We will also perform secondary sample screening for distillation of classification and regression, respectively. We compare the cost of the same sample between the student and teacher models as follows:

$$\begin{cases} \mathcal{A} = \left\{ e \in \mathcal{E} | \mathcal{C}^t_{\text{cls}}(e) < \mathcal{C}^s_{\text{cls}}(e) \right\} \\ \mathcal{B} = \left\{ e \in \mathcal{E} | \mathcal{C}^t_{\text{reg}}(e) < \mathcal{C}^s_{\text{reg}}(e) \right\}. \end{cases} \tag{5}$$

The $\mathcal{E}$ represents the sample set for feature information. The $\mathcal{A}$ contains the samples for mining classification information and the $\mathcal{B}$ includes the samples for mining regression information. $\mathcal{A}^N$ denotes a set of N-types of samples, such as $\mathcal{A}^2 = \{(e_i, e_j) | i \neq j\}$. The $\mathcal{C}^t_{\text{cls}}$, $\mathcal{C}^s_{\text{cls}}$, $\mathcal{C}^t_{\text{reg}}$, and $\mathcal{C}^s_{\text{reg}}$ represent the classification costs and regression costs of the teacher and the student model, respectively.

The whole statistical sample selection module is illustrated in Fig. 4.

C. Implicit Knowledge Mining

The relationship between different distillation samples can reflect the implicit information. More samples provided by extensive objects in RS images bring deeper connection, which is usually ignored by existing distillation approaches. Transferring the relation knowledge from the teacher model to the lightweight student model is contributed to optimizing the model from the implicit perspective. Considering the potency of implicit information, we explore the category relation of the sample pairs and mine the geometry relation of the bounding box groups.

1) Category Relation: Distillation samples usually have close relationships in RS images (such as ship and harbor and plane and airport), this category relation mining will help the student model understand data effectively. Here, we use the Euclidean distance to measure the category relation between samples

$$\psi_D(e_i, e_j) = \frac{1}{\alpha} \| n_i - n_j \|_2 \tag{6}$$

where n_i and n_j denote the classification results of random samples e_i and e_j from $\mathcal{A}$. α is normalization factor for distance. We set the α to be the mean distance between two samples from $\mathcal{A}$ as follows:

$$\alpha = \frac{1}{|\mathcal{A}^2|} \sum_{(e_i, e_j) \in \mathcal{A}^2} \| n_i - n_j \|_2. \tag{7}$$

Then, we adopt the category relation function $\psi_D(\bullet)$ to measure the classification results of sample pairs of the student and teacher models, respectively. After getting the category

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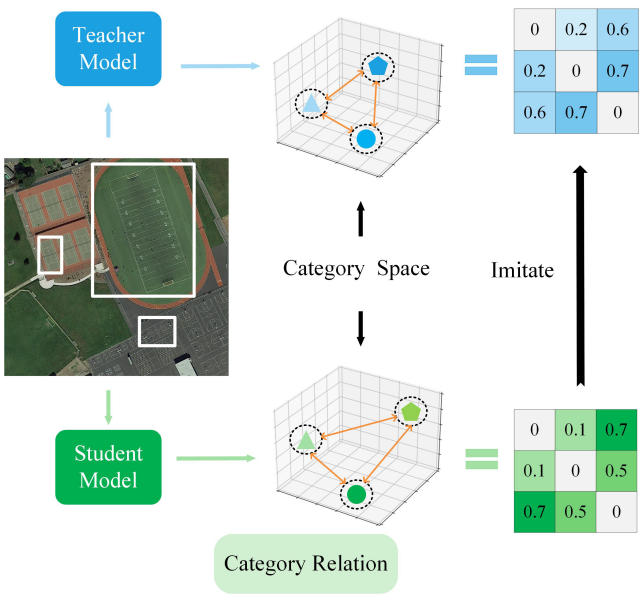

Fig. 5. Illustration of category relation knowledge mining. White rectangles denote statistical samples, and their classification results in category space are acquired by inference of the model. Then, the category relation matrix is calculated by their classification results. The module helps the student to imitate the category relation knowledge of the teacher model.

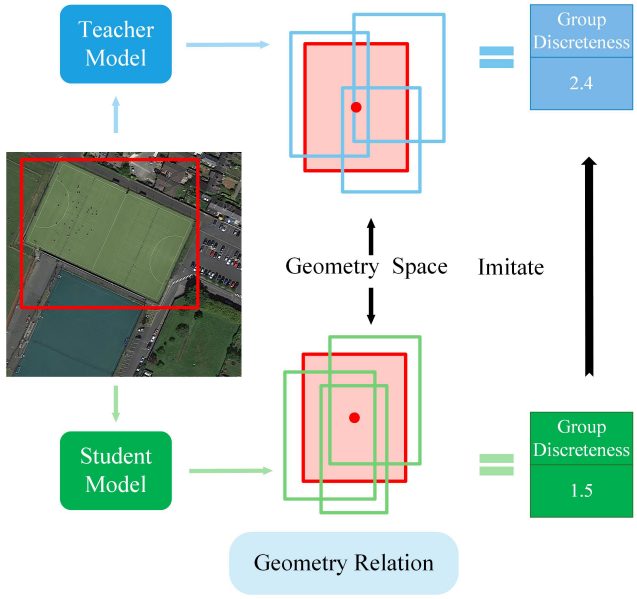

Fig. 6. Illustration of geometry relation knowledge mining. The red rectangle denotes ground truth and the nearby rectangles represent the statistical samples belonging to it. The group discreteness is calculated by their regression results with ground truth. The module helps the student to imitate the geometry relation knowledge of the teacher model.

relation matrix, the category relation distillation loss is defined as follows:

$$\mathcal{L}_C = \sum_{(e_i,e_j)\in\mathcal{A}^2} l_\sigma\left(\psi_D\left(t_i^c, t_j^c\right), \psi_D\left(s_i^c, s_j^c\right)\right) \tag{8}$$

where l_σ is the smooth L1 loss. t_i^c, t_j^c and s_i^c, s_j^c represent the classification results of random samples e_i, e_j of the teacher and student models from the set $\mathcal{A}$.

As shown in Fig. 5, the category relation mining emphasizes the connection between sample pairs, which facilitates the student to study category relation knowledge from the teacher model.

2) Geometry Relation: Massive objects provided by RS images bring more samples, while samples belonging to the same ground truth have a geometry relation, which is usually neglected before. To mine the geometry relation, we first separate the set $\mathcal{B}$ into K groups, K means the number of ground truths in the image.

Hence, a sample group is defined as follows:

$$\mathcal{B}_k = \left\{e \in \mathcal{B} \cap e \in gt^k\right\} \tag{9}$$

where gt^k represents samples that have the same target (ground truth), $k \in [1, K]$.

The geometry relation of a sample group $\mathcal{B}_k$ can be regarded as the discreteness sum of all samples relative to their ground truth. Hence, we first calculate the discreteness of every sample in the set $\mathcal{B}_k$ as follows:

$$\phi_R(p) = 1 - \frac{I(p,gt)}{U(p,gt)} + \frac{E(p,gt) - U(p,gt)}{E(p,gt)} \tag{10}$$

where p is the prediction box of the sample e, $I(p, gt)$ denotes the intersection of the sample prediction box and the corresponding ground truth box, $U(p, gt)$ represents the union of the sample prediction box and the corresponding ground truth prediction box, and $E(p, gt)$ represents the minimum enclosing box of the sample prediction box and the corresponding ground truth prediction box.

The group discreteness is calculated as follows:

$$\psi_R(\mathcal{B}_k) = \frac{1}{\beta}\sum_{e\in\mathcal{B}_k}\phi_R(p) \tag{11}$$

where β is the normalization factor of group discreteness and it is the mean discreteness of every group from $\mathcal{B}$

$$\beta = \frac{1}{|K|}\sum_{\mathcal{B}_k\in\mathcal{B}}\psi_R(\mathcal{B}_k). \tag{12}$$

Then, we adopt the geometry relation function $\psi_R(\bullet)$ to evaluate the group discreteness of the student and teacher models, respectively. Finally, the geometry relation KD is defined as follows:

$$\mathcal{L}_G = \sum_{k=1}^{K} l_\sigma\left(\psi_R\left(\mathcal{B}_k^t\right), \psi_R\left(\mathcal{B}_k^s\right)\right) \tag{13}$$

where l_σ is the smooth $L1$ loss. $\mathcal{B}_k^t$ and $\mathcal{B}_k^s$ represent the sample group of the teacher and student models from the set $\mathcal{B}$.

The geometry relation mining conveys potential knowledge from the perspective of graphic structure in Fig. 6.

Therefore, the total loss of IKM is defined as follows:

$$\mathcal{L}_{\text{imp}} = \lambda_1\mathcal{L}_C + \lambda_2\mathcal{L}_G. \tag{14}$$

We set $\lambda_1 = \lambda_2 = 1$ during the training.

D. Explicit Knowledge Mining

Different from IKM, EKM focuses more on the direct results match between the student and teacher models, which

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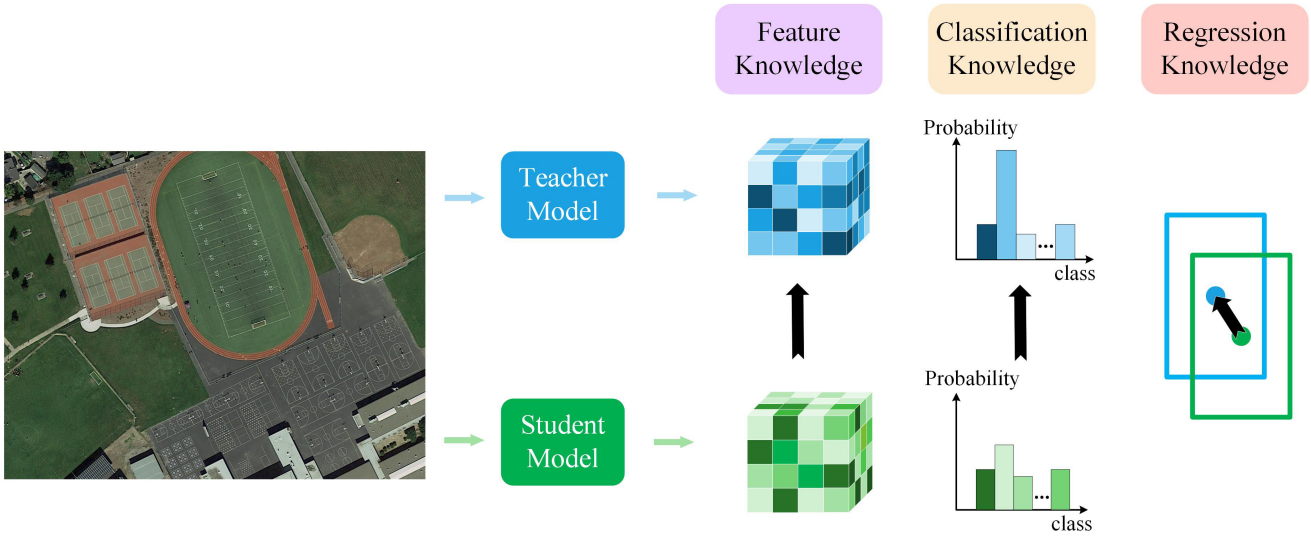

Fig. 7. Illustration of EKM module. The student model imitates the feature, classification, and regression knowledge from the teacher model by different information mining sets.

is shown in Fig. 7. Since the redundancy of the samples, we just transfer the features of the sample in the set $\mathcal{E}$ from the teacher to student models. As for the predictions of the sample, we deliver the knowledge of classification and regression between the student and teacher models in the set of $\mathcal{A}$ and $\mathcal{B}$, respectively, which are obtained by the statistical sample selection module.

1) Feature Knowledge: Feature knowledge contains the key information of the sample, so we let the student model imitate the sample features of the feature pyramid module from the teacher model.

$$\mathcal{L}_f = \frac{1}{|\mathcal{E}|}\sum_{e_i \in \mathcal{E}} \left\| t_i^f - s_i^f \right\|_2^2 \tag{15}$$

where t_i^f and s_i^f represent the features of the sample e_i of the teacher and student models from the set $\mathcal{E}$, respectively.

2) Classification Knowledge: For the classification knowledge, we adopt the $L2$ loss to deliver it as follows:

$$\mathcal{L}_c = \frac{1}{|\mathcal{A}|}\sum_{e_i \in \mathcal{A}} \left\| t_i^c - s_i^c \right\|_2^2 \tag{16}$$

where t_i^c and s_i^c represent the classification results of the sample e_i of the teacher and student models from the set $\mathcal{A}$, respectively.

3) Regression Knowledge: To avoid the issue brought by the direct match of the regression outputs, we just use the GIoU loss to transfer the regression knowledge as follows:

$$\mathcal{L}_r = \frac{1}{|\mathcal{B}|}\sum_{e_i \in \mathcal{B}} \mathrm{GIoU}\left(t_i^r, s_i^r\right) \tag{17}$$

where t_i^r and s_i^r represent the regression results of the sample e_i of the teacher and student models from the set $\mathcal{B}$.

Therefore, the total loss of EKM is defined as follows:

$$\mathcal{L}_{\exp} = \lambda_3 \mathcal{L}_f + \lambda_4 \mathcal{L}_c + \lambda_5 \mathcal{L}_r. \tag{18}$$

We set $\lambda_3 = \lambda_4 = \lambda_5 = 1$ during the training.

Algorithm 1 S^{3}MKM

Input:
$D = \{X, Y\}$ is data for training, X, Y are images and their ground truth, respectively;
$\mathcal{T}$, θ_t are the teacher model and its parameters;
Output:
$\mathcal{S}$, θ_s are the student model and its parameters;
Load the θ_t and initialize the θ_t.
for every batch $D_b = \{X_b, Y_b\}$ **do**
  Obtain the selected samples $\mathcal{A}$ and $\mathcal{B}$ by statistical sample selection module;
  Obtain the features and predictions of teacher model: $\{t^f, t^c, t^r\} \leftarrow \mathcal{T}(X_b, \theta_t)$;
  Obtain the features and predictions of student model: $\{s^f, s^c, s^r\} \leftarrow \mathcal{S}(X_b, \theta_s)$;
  Mine the implicit knowledge: $\mathcal{L}_{imp} = \mathcal{L}_C(t^c, s^c) + \mathcal{L}_G(t^r, s^r)$;
  Mine the explicit knowledge: $\mathcal{L}_{exp} = \mathcal{L}_f(t^f, s^f) + \mathcal{L}_c(t^c, s^c) + \mathcal{L}_r(t^r, s^r)$;
  Calculate the detection loss: $\mathcal{L}_{det}(s^c, s^r, Y_b)$;
  Acquire the whole loss: $\mathcal{L} = \mathcal{L}_{det} + \mathcal{L}_{exp} + \mathcal{L}_{imp}$;
  Update the parameters θ_s of student model $\mathcal{S}$: $\theta_s = argmin_{\theta_s}\mathcal{L}$;
end for
return $\mathcal{S}, \theta_s$.

E. Optimization

In this part, we will introduce the optimization for the distillation framework. We adopt the offline distillation strategy, the teacher model is fixed, and the student model is updated by the dark knowledge of the teacher model and the ground truth in the training phase. The training loss includes a task-specific

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loss (detection loss) and two distillation losses as follows:

$$\mathcal{L} = \mathcal{L}_{\text{det}} + \mathcal{L}_{\text{imp}} + \mathcal{L}_{\text{exp}}. \tag{19}$$

The pipeline of the framework is elucidated in Algorithm 1.

IV. Experiment

In this section, we will first introduce the experimental details, including the dataset and implementation settings. Then, the results and analysis of the ablation experiment are presented. Finally, we will compare our methods with other SOTA distillation approaches.

A. Dataset

1) DIOR: The DIOR dataset [33] is large-scale RS imagery for the task of horizontal object detection. It involves 23 463 images and 192 472 instances. The size of each image is 800 × 800. A total of 11 725 images are chosen as the training set and 11 738 images are selected as the testing set.

2) NWPU VHR-10: Chen *et al.* [34] proposed the NWPU VHR-10 dataset in 2016, which contains ten categories and 800 images. Besides, 650 of them have objects, so we choose 520 images as the training set and 130 images as the testing set. The size of the image is 1000 × 600.

3) DOTA: The DOTA dataset proposed by Xia *et al.* [35] contains 2806 large images and 188 282 instances, which are divided into 15 categories. The size of the image ranges from 800 × 800 to 4000 × 4000, and we crop them into 1024 × 1024 with the overlap of 200 pixels. Half of the cropped images are selected as the training set, 1/6 as the validation set, and 1/3 as the testing set.

B. Implementation Details

1) Model Setting: Both the teacher and student models adopt the classic one-stage model which is introduced in Section III-A. We use the ImageNet pretrained model as the backbone for a better converge.

Teacher Model: We adopt the **ResNet101** as the backbone for the teacher model on the datasets DOTA and DIOR. To avoid overfitting on the NWPU dataset, **ResNet50** is adopted as the backbone.

Student Model: In order to design a lightweight student model, we choose the **ResNet18** as the backbone for the student model on all datasets. For the DOTA and DIOR datasets, the number of Feature Pyramid Networks (FPN) channels is reduced to half. But for the NWPU dataset, we reduce its FPN channels to a quarter of the original.

2) Training and Testing Strategy: The detail of training strategies for various datasets is shown in Table I. We use stochastic gradient descent (SGD) to optimize the parameters of the student model. We set the warm-up learning rate to 0.001 with the first 500 rounds. The weight decay and momentum are set as 0.0001 and 0.9.

In the test stage, the IoU threshold of nonmaximum suppression is 0.4 on DOTA and DIOR, and 0.6 on NWPU VHR-10. We perform our experiments on an NVIDIA 3090 GPU with Pytorch 1.7.

TABLE I
Training Strategy

Dataset	Image Size	Batch Size	Lr	Epoch
DOTA	1024×1024	8	0.01	24 (20,22)
NWPU VHR-10	1000×600	4	0.005	40 (30)
DIOR	800×800	8	0.01	24 (20,22)

3) Evaluation Metric: We adopt the mean average precision (mAP) to evaluate the performance of our model on the dataset. The mAP is related to TP, FP, FN, Precision, and recall. A true positive (TP) means the overlap ratio between the ground true box and the bounding box predicted by a detection result surpasses a certain threshold. Otherwise, it will be seen as a false positive (FP). False negative (FN) represents a ground truth that has no matching prediction results. Therefore, we will obtain the precision and recall as follows:

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{20}$$

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{21}$$

We get different precision and recall according to the various confidence, then a P-R curve is drawn by them. AP can be calculated by the area below the P-R curve. Finally, we get the mAP by calculating the average value of all categories' AP.

We evaluate the model efficiency and lightweight performance by adopting frames per second (FPS), floating-point of operations (FLOPs), and amount of network parameters (Params).

C. Ablation Studies

In this part, we will conduct ablation studies to verify the effectiveness of our proposed approach.

1) Effectiveness of Multivariate Knowledge Mining: As presented in Section III-C and III-D, we discuss two different knowledge mining methods. The EKM module contains the feature, classification, and regression knowledge. First, as shown in Table II, we adopt EKM with various knowledge combinations. No matter whether one kind of knowledge, two kinds of knowledge, or three kinds of knowledge, the performance of the lightweight model can be efficiently promoted. The classification knowledge can improve the highest 4.59% accuracy among all the single knowledge of EKM. Moreover, the combination of feature and classification knowledge will achieve the best improvement of 4.65% in the two-tuple of knowledge. We obtain the promotion of 4.96% when we unite all the knowledge of EKM. Besides, IKM is also used to mine the relation knowledge among samples. The category relation can strengthen the connection of samples, which promotes 3.39% accuracy. The geometry relation aggregates the samples which belong to the same ground truth by group discreteness; hence, it improves 4.86% performance. When combining the category relation and geometry relation, our method transcends the baseline of 5.06%. Finally, multivariate knowledge mining

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TABLE II

ABLATION STUDY OF MULTIVARIATE KNOWLEDGE MINING. THE EKM REPRESENTS THE EXPLICIT KNOWLEDGE MINING MODULE AND THE IKM DENOTES THE IMPLICIT KNOWLEDGE MINING MODULE. MKM MEANS OUR MULTIVARIATE KNOWLEDGE MINING

Method	Feature	Classification	Regression	Category	Geometry	mAP	ΔAP50
teacher	-	-	-	-	-	71.65	-
student	-	-	-	-	-	63.80	-
EKM	√	-	-	-	-	67.21	+3.41
	-	√	-	-	-	68.39	+4.59
	-	-	√	-	-	68.04	+4.24
	√	√	-	-	-	68.45	+4.65
	√	-	√	-	-	67.73	+3.93
	-	√	√	-	-	67.42	+3.62
	√	√	√	-	-	68.76	+4.96
IKM	-	-	-	√	-	67.19	+3.39
	-	-	-	-	√	68.66	+4.86
	-	-	-	√	√	68.86	+5.06
MKM	√	√	√	√	√	**69.00**	**+5.20**

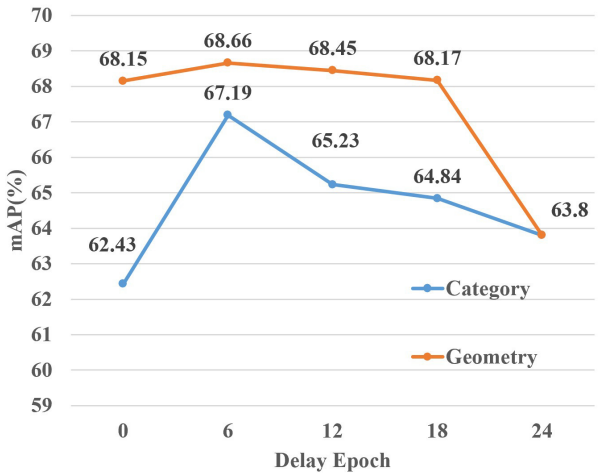

Fig. 8. Analysis of delay strategy. We add the IKM at four stages (0, 6, 12, and 18) during the training. The 24 means the training without IKM.

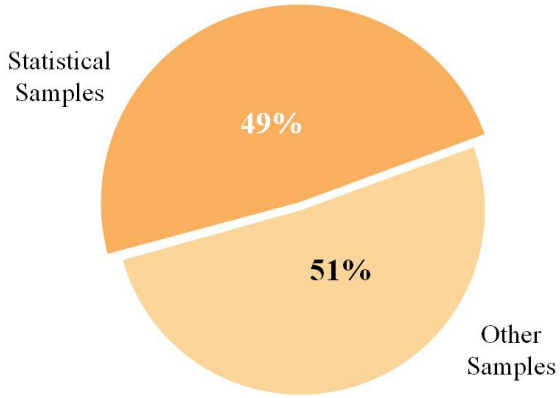

Fig. 9. Statistical sample selection module chooses the 49% optimal sample for distillation from all the positive samples.

TABLE III

ABLATION STUDY OF STATISTICAL SAMPLE SELECTION MODULE. GMM REPRESENTS THE SAMPLE SELECTED BY GMM, BGM DENOTES THE SAMPLE SELECTED BY BAYESIAN GMM, AND POS REFERS TO ALL THE POSITIVE SAMPLES

Method	Pos	BGM	GGM	mAP	ΔAP50
student	-	-	-	63.80	-
distillation	√	-	-	67.82	+4.02
	-	√	-	68.07	+4.27
	-	-	√	**69.00**	**+5.20**

can improve the performance of the lightweight model validly as illustrated in the last row of Table II (5.20% promotion).

2) Effectiveness of Delay Strategy: Different from EKM, IKM emphasizes more about the potential relationship. During the prophase of training, the model is underfitting so that the relation mining will result in the deviation of optimizing. Therefore, we design a delay strategy, which adds IKM after certain epochs of the training. As shown in Fig. 8, no matter the adding in the prophase, or the anaphase, the performance is worse than the metaphase of the training. Hence, we adopt the IKM in the training metaphase.

3) Analysis of Statistical Sample Selection: To validate the effectiveness of our statistical sample selection, we first calculate the percent of statistical samples among all the positive samples. As shown in Fig. 9, our method selects 49% samples for distillation on average for every epoch. Then, our method is compared with a common method (Pos) that uses all the positive samples for distillation. "Pos" can be viewed as the baseline of the sample selection. Besides, we also add the Bayesian BGM to replace the GMM for comparison. The BGM is initialized as the default setting and the GMM is initialized as follows. weights_init = [0.5, 0.5], means_init = [mix_cost, max_cost], and precisions_init = [1, 1]. We test

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TABLE IV
ABLATION STUDY OF HETEROGENEOUS STUDENT NETWORKS

Method	S^3MKM	mAP	ΔAP50
teacher (ResNet101)	-	71.65	-
MobileNetV2	-	56.91	-
	√	**58.03**	**+1.12**
RegNetX-800mf	-	67.75	-
	√	**69.84**	**+2.09**

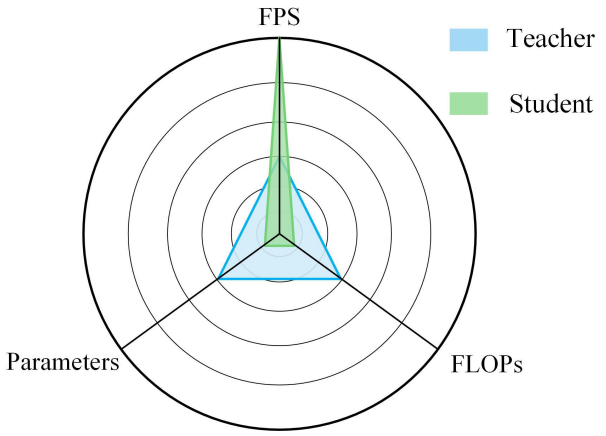

Fig. 10. Comparisons of the teacher and lightweight student model in terms of FPS, Parameters (M), and FLOPs (G).

two methods many times and adopt the max mAP as their results. Compared with the BGM, the GMM has a better manual initialization, which means better results. Table III represents the value of the statistical sample, which achieves the highest accuracy. The optimal samples for distillation will better improve the performance of the lightweight model.

4) Analysis of Heterogeneous Student Network: To verify the generalization of our proposed method, we use two heterogeneous student backbone, such as **MobileNetV2** [6] and **RegNetX-800mf** [36]. As shown in Table IV, whether the backbone is the MobileNetV2 or RegNetX-800mf, they both get a performance gain even if the student backbone is heterogeneous with teacher.

5) Analysis of Lightweight: The student model designed by us is very lightweight, as shown in Fig. 10. Its inference speed (**42.7 FPS**) is two times of the teacher model, but its FLOPs (**41.6 G**) and parameters (**13.1M**) are 23.9%, 25.7% of the teacher model, respectively.

6) Qualitative Analysis: As illustrated in Fig. 11, we first analyze the AP distribution of 15 classes. From the box plot, we can see the distribution of baseline is more discrete, which has a bigger range. However, the results of our method are more concentrated, which means ours is more robust for each class detection.

Besides, Fig. 12 shows that the multivariate knowledge mining facilitates the lightweight model to capture more accurate feature information, our method focuses more on the object, rather than backgrounds compared with the baseline.

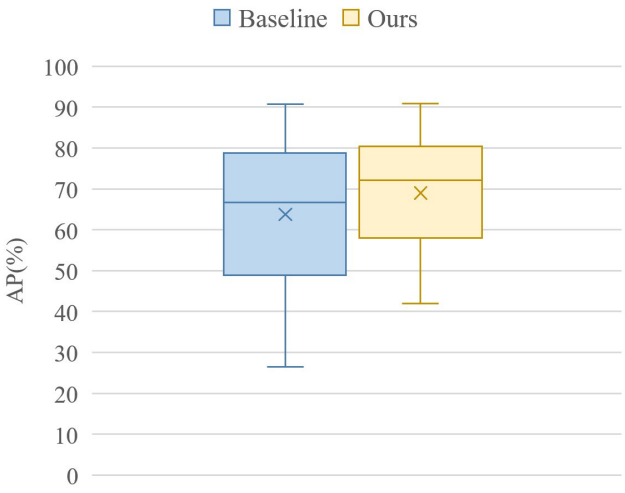

Fig. 11. Box-plot shows the AP distribution of 15 classes (baseline and our method). The lower and higher short lines denote the minimum and maximum, respectively. The x represents the mean.

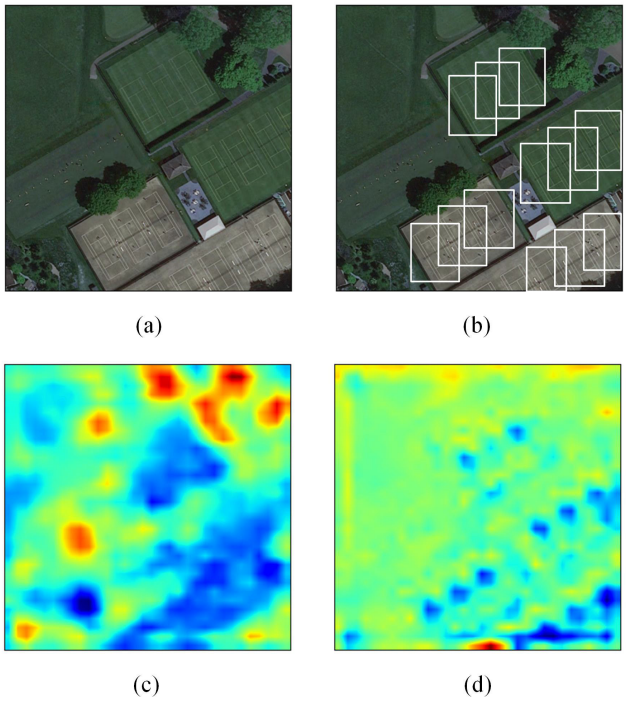

Fig. 12. Visualization of features. (a) Input image. (b) Ground truth. (c) Baseline. (d) Our method.

Moreover, we also test the effect of the varying illumination area. Fig. 13 (a)–(d) shows the detection results for the category "chimney." It can be seen that although the lighting conditions are changing, our algorithm can still detect "chimney" effectively.

Finally, we will discuss the radiometric contrast of the object with the background. We choose two similar pictures for evaluation, both of them have the category of stadium and ground track field. The visualization comparisons are presented in Fig. 14(a) and (d) are original pictures. The radiometric

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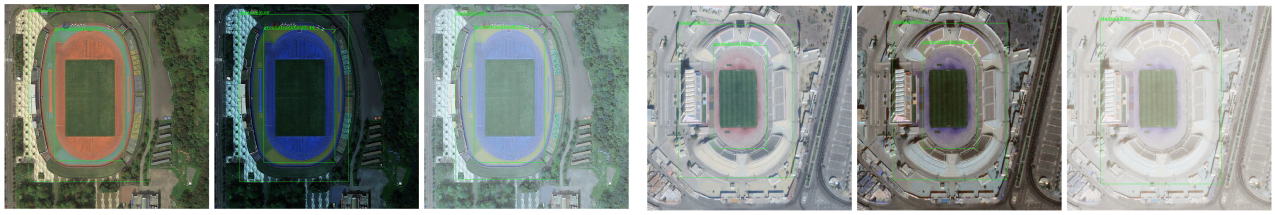
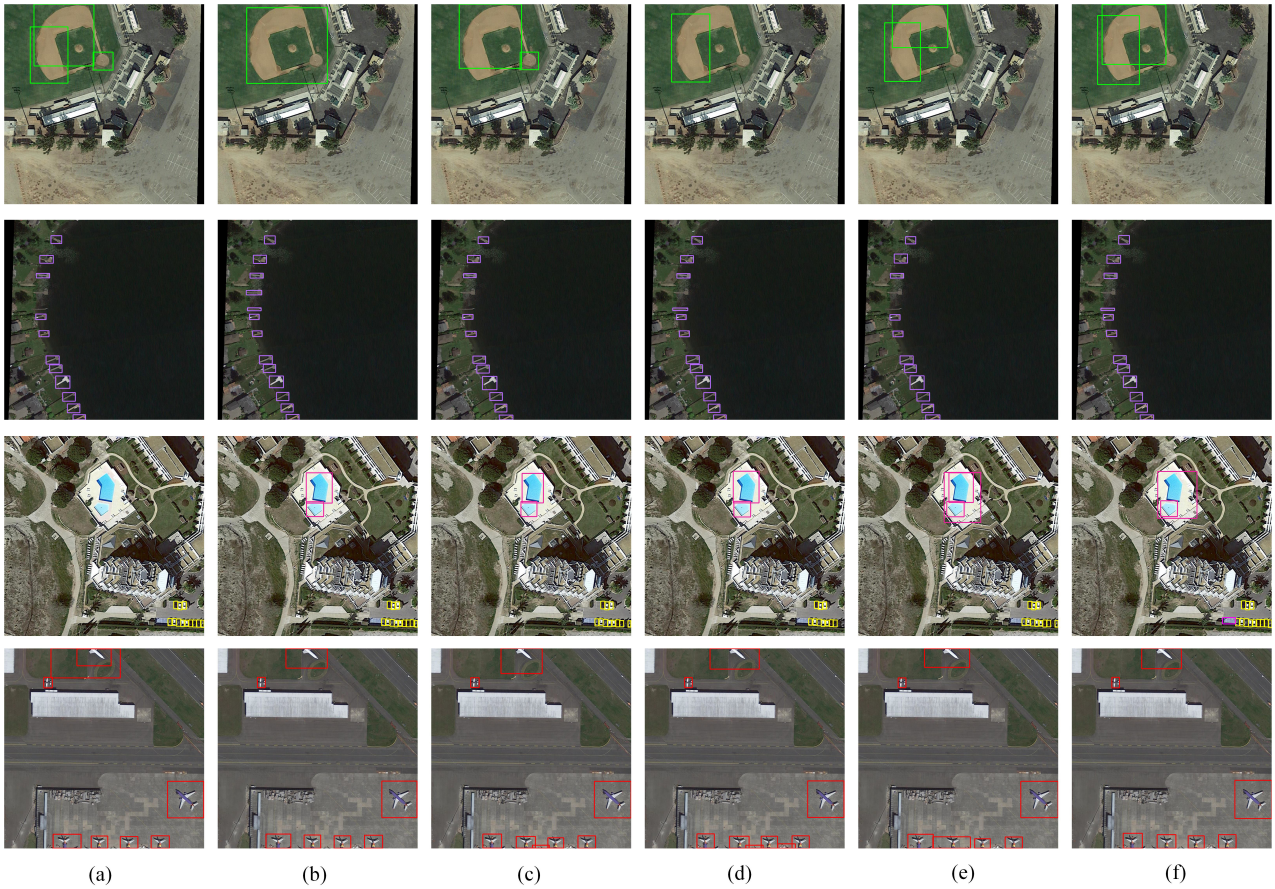
(a) (b) (c) (d)

Fig. 13. Detection visualization of the different illumination conditions. (a) and (b) Weak illumination conditions. (c) and (d) Strong illumination conditions.



(a) (b) (c) (d) (e) (f)

Fig. 14. Detection visualization of the varying radiometric contrast of the object with the background. (a) and (d) Original pictures. (b) and (e) Adjust the contrast by gamma = 2. (c) and (f) Adjust the contrast by gamma = 0.5.



(a) (b) (c) (d) (e) (f)

Fig. 15. Visualization of comparison between our method with other methods. (a) Baseline. (b) Ours. (c) KD. (d) FitNet. (e) FGFI. (f) TAR.

contrast of the object with the background is weak in Fig. 14(a). But the radiometric contrast of the object with the background is strong in Fig. 14(d). Fig. 14(b) and (e) adjusts the contrast by gamma = 2, and Fig. 14(c) and (f) adjusts the contrast by gamma = 0.5. The visualization results present our algorithm's robustness for radiometric contrast.

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TABLE V
COMPARISON WITH SOTA DISTILLATION APPROACHES ON DOTA, NWPU VHR-10, AND DIOR DATASETS

	DOTA		NWPU VHR-10		DIOR	
method	mAP	ΔAP50	mAP	ΔAP50	mAP	ΔAP50
teacher	71.65	-	93.21	-	71.7	-
student	63.80	-	86.56	-	65.2	-
DeFeat [30]	64.11	+0.31	86.92	+0.36	67.7	+2.5
KD [9]	65.80	+2.00	86.82	+0.26	67.2	+2.0
FGFI [15]	66.25	+2.45	88.69	+2.13	69.5	+4.3
FitNet [10]	66.30	+2.50	86.85	+0.29	69.2	+4.0
RKD [12]	67.19	+3.39	90.20	+3.64	67.3	+2.1
TAR [16]	68.09	+4.29	90.22	+3.66	69.5	+4.3
ARSD [31]	68.28	+4.48	90.92	+4.36	70.1	+4.9
S^3MKM (ours)	**69.00**	**+5.20**	**92.90**	**+6.34**	**70.2**	**+5.0**

TABLE VI
COMPARISON WITH CLASSIC DETECTORS ON DOTA, NWPU VHR-10, AND DIOR DATASETS

	DOTA				NWPU VHR-10				DIOR			
method	mAP (%)	FPS (f/s)	Params (M)	FLOPs (G)	mAP (%)	FPS (f/s)	Params (M)	FLOPs (G)	mAP (%)	FPS (f/s)	Params (M)	FLOPs (G)
Faster R-CNN [4]	60.64	7.0	60.19	289.25	77.80	25.4	41.17	127.70	54.1	16.8	60.21	182.02
RetinaNet [19]	50.39	14.0	55.39	293.36	89.40	26.3	36.29	123.27	65.7	18.2	55.49	180.62
YOLOv3 [37]	60.00	13.0	61.63	198.92	88.30	31.2	61.57	121.27	57.1	14.2	61.95	122.22
FCOS [38]	67.72	18.3	31.57	202.15	89.65	24.4	31.86	116.63	67.6	25.4	31.88	123.51
ATSS [20]	66.84	26.6	18.97	156.01	90.50	37.4	18.96	89.90	67.7	26.7	18.98	95.50
GFL [39]	66.53	23.8	19.13	159.18	88.80	34.7	19.13	91.73	68.0	36.5	19.13	97.43
MobileNetV2 [6]	56.91	22.5	10.30	124.24	76.9	30.2	10.29	71.49	58.2	30.1	10.32	76.1
ShuffleNet [40]	57.73	22.2	12.11	142.60	83.0	25.9	12.10	82.17	61.3	25.0	12.12	87.31
S^3MKM (ours)	**69.00**	**36.7**	**13.08**	**68.03**	**92.90**	**47.1**	**11.57**	**26.65**	**70.2**	**42.7**	**13.10**	**41.60**

D. Comparisons With SOTA Distillation Methods

In this section, we compare our method with other SOTA distillation methods. We adopt the same teacher and student models introduced in Section IV-B1. The results are illustrated in Table V. The experiments on three datasets (DOTA, NWPU VHR-10, and DIOR) demonstrate the effectiveness of our method and we achieve state-of-the-art performance.

We first migrate two methods used in the task of classification. The KD [9] focuses on the classification knowledge, and the FitNet [10] pays attention to the features knowledge. They both improve the accuracy of the lightweight model to some extent. Then, we compare three methods designed for the detection task. The FGFI [15] imitates the features knowledge close to the objects in the feature map of the teacher model. TAR [30] chooses the ground truth region as feature knowledge and adds the classification and regression knowledge of the positive samples. The Defeat [30] introduces dark knowledge of the background besides the foreground. They also bring some promotion. The RKD [12] adopts the relation knowledge between images as the key information. Finally, ARSD [31] designs an adaptive distillation framework for RS images. Different from their monotonous knowledge mining and static sample selection, our method dynamically selects the optimal multiscale RS distillation samples without complex hand-craft hyperparameters and synchronizes the training phase. The multivariate knowledge mining proposed by us delivers the various knowledge sufficiently, which validly strengthens the performance of the lightweight model. And we also compare the visualization of our method with some other approaches, as shown in Fig. 15. The visualization results certify our method can effectively improve the accuracy of the lightweight model, which has fewer false alarms, misclassification, and more accurate location.

E. Comparison With the Classic Detectors

In this part, to assess the accuracy of our lightweight model, we compare it with other classic heavy detectors (faster R-CNN [4], RetinaNet [19], YOLOv3 [37], Fully Convolutional One-Stage Object Detection (FCOS) [38], Adaptive Training Sample Selection (ATSS) [20], and GFL [39]) as illustrated in Table VI. Our lightweight model has the optimal performance, the fastest inference speed, and the smallest parameters and FLOPs. The results prove

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the potency of the lightweight model, which means the heavy detectors have so much redundancy and unnecessary parameters. Besides, we also compare our model with other lightweight detectors, such as MobileNetV2 [6] and ShuffleNet [40]. We replace the backbone of the baseline with the MobileNetV2 or ShuffleNet. Both of them adopt depthwise separable convolutions which make them have larger FLOPs. As shown in Table VI, although their model parameters are close to ours, their inference speeds are slower than ours, and accuracies are about 10% less than ours. Therefore, our model has a better tradeoff between performance and lightweight.

V. Conclusion

In this article, we have proposed a general distillation framework (S^3MKM) to optimize lightweight detectors in RS imagery. The statistical sample selection module is designed to dynamically determine the optimal multiscale RS distillation sample without complex hand-craft hyperparameters. The module models all the positive sample selection costs as a probability distribution, maximizes its likelihood by the EM algorithm, and separates them by a low-cost scheme to obtain the statistical samples. Therefore, the sample for distillation and detection task training are synchronized, which contributes to the lightweight model optimization. Furthermore, the multivariate knowledge mining module transfers the explicit and implicit knowledge of the statistical samples, which helps the lightweight model optimize the parameters better. The experiments on three datasets (DOTA, NWPU VHR-10, and DIOR) demonstrate that our method achieves state-of-the-art performance. Besides, the performance of the lightweight model optimized by our S^3MKM surpasses other heavy classic detectors, which shows its potency. In the future, we will focus more on lightweight model optimization and combine the KD with other methods.

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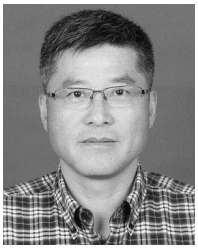

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